

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Lecture 2: Momentum Conservation and Medium Stress

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Overview

Matter and medium contributions are combined so that the total stress-energy tensor obeys a local conservation law.

1 Physical assumptions

All coordinate conditions and boundary conditions are stated before a parameter is fitted.

$$\partial_\mu (T^{\mu\nu} + S^{\mu\nu}) = 0$$

2 Field equations

The first two displayed relations establish the notation used in the derivation.

$$P_\Omega^i = \int_\Omega (T^{0i} + S^{0i}) d^3x$$

3 Observable limit

Weak-field, static, and point-source limits are checked independently.

$$S^{\mu\nu} = \varepsilon_1 A^{\mu\nu} + \varepsilon_2 B^{\mu\nu}$$

4 Parameter dependence

Parameter combinations are compared through derivatives of the reference observables.

5 Example

The calculation is repeated for one analytic case and one tabulated data point.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Einstein–Mythos paper E, §§3–7
- Hall, ch. 2
- SGT9 conservation block C-01–C-06